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Executive Summary

Quantitative Software Engineer & Automated Trading Architect. Creator of AURA (24/7 automated spot & equity execution engine). Expert in walk-forward optimization, successive halving, multi-layer risk management, PAXG defensive overlays, and high-throughput Python/C++ architectures.

Tailored Competencies for Deutsche Bank (Quantitative Developer, Trading and Client Controls (TaCC))

- Core Technical Stack: Python 3.11+ / AsyncIO, C++ High-Performance Data Processing, pandas / NumPy / SciPy, CCXT Multi-Exchange Integration, AURA Proprietary Execution Architecture, Walk-Forward Optimization & Successive Halving, Multi-Layer Risk Management (Dynamic SL/TP, Volatility Regime Detection), PAXG Defensive Asset Allocation Overlays
- Product Showcase: EduTrace (https://edutrace.net) & AURA (https://www.auratrading.org/)

Professional Experience

Software Manager | NISO TECHNOLOGY CO

May 2021 - Present | Izmir, TR

Directed a multidisciplinary engineering organization of 30 software engineers across 4 team leads, delivering mission-critical embedded software for EV powertrain systems including ECUs, traction inverters, and on-board chargers (OBCs).

- Scaled organizational capability by establishing a team-lead layer, defining ownership boundaries, and running structured performance reviews and mentorship programs - reducing manager single points of failure and enabling parallel delivery across subsystems.
- Enforced ISO 26262 and SPICE compliance across BSW and ASW layers developed on FreeRTOS and other RTOS platforms, ensuring functional safety traceability from requirements through integration.
- Deployed CI/CD pipelines and automated tooling that compressed release cycle times and shifted regression detection left, reducing late-stage integration failures.
- Led UN R155/R156 cybersecurity standard implementation activities, aligning software development processes with vehicle cybersecurity management system (CSMS) requirements.
- Designed and deployed C-based middleware solutions for high-reliability embedded applications, and contributed directly to diagnostic and safety-critical algorithms for powertrain systems.
- Instituted Git-based version control strategies with branching policies that enforced traceability and streamlined cross-team collaboration across a large, concurrent codebase.
- Drove data-driven project management by deploying analytics tooling to surface velocity, defect density, and milestone risk metrics to stakeholders.

- Managed resource allocation and risk mitigation across competing program milestones, acting as the primary interface between software, hardware, and systems engineering teams.

Team Leader | ECEMTAG

November 2017 - May 2021 | Istanbul, TR

Led design and validation of model-based control algorithms for ECUs and TCUs across automotive and defense programs.

- Introduced automated testing frameworks and HIL simulation environments that raised software verification coverage and reduced physical prototype dependency.
- Developed safety and redundancy functions in compliance with military standards, coordinating requirements directly with defense customers.
- Acted as cross-functional integration lead, aligning software and hardware milestones across embedded, systems, and validation teams using JIRA-based program tracking.
- Contributed to E/E architecture process development and rollout, including tool qualification and development workflow standardization.

Lead Researcher | TUBITAK (Scientific and Technological Research Council of Turkey)

December 2016 - November 2017 | Kocaeli, TR

- Directed automotive and defense research programs focused on advanced control systems and diagnostics.
- Developed a product-level tool for data dictionary generation, calibration, and MiL (Model-in-Loop) testing, accelerating algorithm validation workflows.
- Characterized IPMSM performance in motoring and regenerative modes on a dynamometer test bench.

Senior Systems Engineer | TUMOSAN

January 2016 - December 2016 | Istanbul, TR

- Developed automatic shift-selection logic using Simulink Stateflow for a hybrid drivetrain platform.
- Designed traction control algorithms for slip-slide protection under variable surface conditions.
- Optimized fuel consumption and battery usage profiles using multi-objective cost functions.

Systems Engineer | METRO CO

February 2013 - January 2016 | Kocaeli, TR

- Built a longitudinal vehicle model and derived drivetrain power and torque requirements from first principles.
- Developed a full PMSM driver including Maximum Torque Per Ampere (MTPA) and flux-weakening control algorithms.
- Automated PID tuning via phase and gain margin calculation, and conducted HIL validation using dSpace MABX-I.

Researcher | Iowa State University - Bio-Electric Laboratory

February 2011 - January 2013 | Ames, IA

- Supervised advanced research on transcranial magnetic stimulation (TMS) and electric field modeling.
- Instructed electric machines coursework to sophomore engineering students.
